

Digital Signal Processing (DSP) - Session 3

Discrete-Time Signals and Systems

Analysis of Discrete-Time Linear Time-Invariant Systems

We aim to design **Linear Time-Invariant (LTI) systems** that are both **stable** and **causal**.

There are two basic methods for analyzing the response of a linear system to a given input signal:

- **Direct Solution:** Based on the direct solution of the input-output (difference) equation for the system.
- **Decomposition:** Decomposing the input signal into a sum of elementary signals, selected so that the response of the system to each component is easily determined.

Resolution of a Discrete-Time Signal into Impulses

A discrete-time signal $x(n)$ can be resolved into a weighted sum of unit sample (impulse) sequences. Each multiplication of the signal $x(n)$ by a unit impulse at a specific delay k picks out the single value $x(k)$.

$$x(n) = \sum_{k=-\infty}^{\infty} x(k)\delta(n-k)$$

Response of LTI Systems: The Convolution Sum

Given a LTI system $\mathcal{T}$:

- The system's response to a unit sample $\delta(n)$ is the **impulse response**.

$$h(n) = \mathcal{T}[\delta(n)]$$

- By the property of time-invariance, the response to a delayed impulse $\delta(n-k)$ is $h(n-k)$.
- By the property of linearity (superposition), the total output $y(n)$ to an arbitrary input $x(n)$ is the sum of the weighted responses to each impulse component:

$$y(n) = \mathcal{T}[x(n)] = \mathcal{T}\left[\sum_{k=-\infty}^{\infty} x(k)\delta(n-k)\right] = \sum_{k=-\infty}^{\infty} x(k)\mathcal{T}[\delta(n-k)] = \sum_{k=-\infty}^{\infty} x(k)h(n-k)$$

- The impulse response $h(n)$ completely characterizes the LTI system, as the output can be determined for any input using this transformation.

The operation

$$\sum_{k=-\infty}^{\infty} x(k)h(n-k)$$

is called the **convolution** of $x(n)$ and $h(n)$, often denoted as

$$x(n) * h(n)$$

Properties of Convolution

Convolution satisfies several mathematical laws:

- **Commutative Law:**

$$x(n) * h(n) = h(n) * x(n)$$

- **Associative Law:**

$$[x(n) * h_1(n)] * h_2(n) = x(n) * [h_1(n) * h_2(n)]$$

- **Distributive Law:**

$$x(n) * [h_1(n) + h_2(n)] = x(n) * h_1(n) + x(n) * h_2(n)$$

- **Identity Property:**

$$x(n) * \delta(n) = x(n)$$

- **Shifting Property:**

$$x(n) * \delta(n - k) = x(n - k)$$

Steps for Computing Convolution

The process of evaluating the convolution sum involves four distinct operations:

1. **Folding:** Fold $h(k)$ about $k = 0$ to obtain $h(-k)$.
2. **Shifting:** Shift the folded sequence by n to the right (if $n > 0$) or left (if $n < 0$) to obtain $h(n - k)$.
3. **Multiplication:** Multiply $x(k)$ by $h(n - k)$ to obtain the product sequence.
4. **Summation:** Sum all values of the product sequence to obtain the output value $y(n)$.

[!EXAMPLE] *Example 3.2*

The impulse response of a linear time-invariant system is

$$h(n) = \{1, \underset{\uparrow}{2}, 1, -1\}$$

Determine the response of the system to the input signal

$$x(n) = \{\underset{\uparrow}{1}, 2, 3, 1\}$$

Solution

1. **Define Input Sequences:**

- **Input signal:**

$$- x(k) = \{\underset{\uparrow}{1}, 2, 3, 1\}, \text{ for } k = 0, 1, 2, 3$$

- **Impulse response:**

$$- h(k) = \{1, \underset{\uparrow}{2}, 1, -1\}, \text{ for } k = -1, 0, 1, 2$$

2. Convolution Sum Formula:

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k)$$

3. Calculation:

$$\begin{aligned} y(-1) &= x(0)h(-1) = (1)(1) = 1 \\ y(0) &= x(0)h(0) + x(1)h(-1) = (1)(2) + (2)(1) = 4 \\ y(1) &= x(0)h(1) + x(1)h(0) + x(2)h(-1) = (1)(1) + (2)(2) + (3)(1) = 8 \\ y(2) &= x(0)h(2) + x(1)h(1) + x(2)h(0) + x(3)h(-1) = (1)(-1) + (2)(1) + (3)(2) + (1)(1) = 8 \\ y(3) &= x(1)h(2) + x(2)h(1) + x(3)h(0) = (2)(-1) + (3)(1) + (1)(2) = 3 \\ y(4) &= x(2)h(2) + x(3)h(1) = (3)(-1) + (1)(1) = -2 \\ y(5) &= x(3)h(2) = (1)(-1) = -1 \end{aligned}$$

4. Final Response Sequence:

$$y(n) = \{1, \underset{\uparrow}{4}, 8, 8, 3, -2, -1\}$$

[!EXAMPLE] Example 3.3

Determine the output $y(n)$ of a relaxed linear time-invariant system with impulse response

$$h(n) = a^n u(n), \quad |a| < 1$$

when the input is a unit step sequence, that is,

$$x(n) = u(n)$$

Solution

5. **Define the sequences:** Both the impulse response $h(n)$ and the input signal $x(n)$ are infinite-duration sequences.
6. **Select the convolution form:** The convolution is computed using the form where the input sequence $x(k)$ is folded and shifted:

$$y(n) = \sum_{k=-\infty}^{\infty} x(n-k)h(k)$$

7. **Evaluate for $n < 0$:** When the shift n is negative, there is no overlap between the non-zero values of $x(n-k)$ and $h(k)$. Consequently, the product sequences consist of all zeros, so $y(n) = 0$ for $n < 0$.
8. **Evaluate for $n \geq 0$:** Compute the sum of the product sequences for successive values of n :
 - For $n = 0$: $y(0) = 1$.
 - For $n = 1$: $y(1) = 1 + a$.
 - For $n = 2$: $y(2) = 1 + a + a^2$.
9. **Generalize the summation:** For any $n > 0$, the output is the sum of a geometric series:

$$y(n) = 1 + a + a^2 + \cdots + a^n = \sum_{k=0}^n a^k$$

10. **Simplify the result:** Applying the geometric series formula, the output for $n \geq 0$ is:

$$y(n) = \frac{1 - a^{n+1}}{1 - a}$$

11. **Identify steady-state response:** Since $|a| < 1$, as n approaches infinity, the output reaches a final value of:

$$y(\infty) = \frac{1}{1 - a}$$

Causality and Stability Conditions

For an LTI system to be practically realizable and useful, it must satisfy specific conditions regarding its impulse response:

- **Causality:** An LTI system is causal if and only if its impulse response is zero for negative time:

$$h(n) = 0, \quad n < 0$$

- **Stability (BIBO):** An LTI system is Bounded-Input Bounded-Output (BIBO) stable if and only if its impulse response is **absolutely summable**:

$$S_h = \sum_{k=-\infty}^{\infty} |h(k)| < \infty$$

Systems with Finite-Duration and Infinite-Duration Impulse Response (FIR and IIR)

Finite-Duration Impulse Response (FIR)

$$h(n) = 0, \quad n < 0, n \geq M$$

$$y(n) = \sum_{k=0}^{M-1} h(k)x(n-k)$$

- **Description:** The impulse response is zero outside of a finite time interval.
- **Interpretation:** The output $y(n)$ at any time is a **weighted linear combination** of the most recent M input signal samples, specifically $x(n), x(n-1), \dots, x(n-M+1)$.
- **Memory:** The system acts as a **window** that only views the most recent M samples and “forgets” all prior inputs, giving it a **finite memory** of length M .
- **Implementation:** FIR systems are easily implemented directly as suggested by the convolution sum, requiring additions, multiplications, and a **finite number of memory locations**. They are typically realized using **nonrecursive** structures.

Infinite-Duration Impulse Response (IIR)

$$y(n) = \sum_{k=0}^{\infty} h(k)x(n-k)$$

- **Description:** The impulse response has an infinite duration.

- **Interpretation:** The output is a weighted linear combination of the present and **all past input samples**.
- **Memory:** Because the output depends on the entire history of the input signal, the system is said to have **infinite memory**.
- **Implementation:** Direct implementation via the convolution summation is **practically impossible** because it would require infinite memory, multiplications, and additions.
- **Realization:** Instead of the convolution sum, IIR systems are more practically and efficiently implemented using **difference equations** and **recursive structures**. These structures use feedback loops to express the current output in terms of past output values as well as current and past inputs.

Recursive and Nonrecursive Discrete-Time Systems

- **Nonrecursive Systems**
 - **Definition:** A system is nonrecursive if its output $y(n)$ at any time n depends only on the present and past values of the input, such that $y(n) = F[x(n), x(n-1), \dots, x(n-M)]$.
 - **Characteristics:** Causal **FIR (Finite-Duration Impulse Response)** systems are inherently nonrecursive.
 - **Computation:** The output can be computed in any order (e.g., $y(200)$ can be found without first finding $y(199)$).
- **Recursive Systems**
 - **Definition:** A system is recursive if the current output $y(n)$ is a function of past output values (e.g., $y(n-1), y(n-2)$) in addition to current and past inputs.
 - **Feedback Loop:** These systems are characterized by a **feedback loop** that contains at least one delay element, which is essential for the system to be practically realizable.
 - **Computation:** Outputs must be computed sequentially in order (e.g., $y(0), y(1), y(2), \dots$).
 - **Initial Conditions:** The response of a recursive system is not uniquely determined by the input alone; it also depends on the **initial conditions** (the state of the system's memory).

Implementation of Discrete-Time Systems

- **Linear Constant-Coefficient Difference Equations:** These are the primary means for implementing linear time-invariant (LTI) systems.
- **Direct Form I:** This structure uses separate sets of delay elements (memory) for the input signal and the output signal. It can be viewed as a cascade of a nonrecursive system followed by a recursive system.
- **Direct Form II:** By interchanging the order of the recursive and nonrecursive parts, the two sets of delays can be merged into a single set.
 - **Canonic Form:** Because it requires the **minimum number of memory locations** ($\max M, N$), it is often referred to as the canonic form.
- **Modular Realization:** Higher-order systems are typically implemented by cascading **second-order sections** as basic building blocks.

Correlation of Discrete-Time Signals

- **Purpose:** Correlation measures the degree to which two signals are **similar** to extract information for applications like radar, sonar, and digital communications.
- **Crosscorrelation**
 - **Formula:** For two real sequences $x(n)$ and $y(n)$, the crosscorrelation is

$$r_{xy}(l) = \sum_{n=-\infty}^{\infty} x(n)y(n-l)$$

- **Non-Commutative:** Unlike convolution, crosscorrelation is generally **not commutative**; changing the order of indices yields $r_{xy}(l) = r_{yx}(-l)$, which is a folded version of the original sequence.
 - **Relation to Convolution:** It can be computed using a convolution program by folding one of the sequences first: $r_{xy}(l) = x(l) * y(-l)$.
- **Autocorrelation**
 - **Definition:** This occurs in the special case where a signal is correlated with itself ($y(n) = x(n)$).
 - **Formula:**

$$r_{xx}(l) = \sum_{n=-\infty}^{\infty} x(n)x(n-l)$$

- **Properties:** The autocorrelation sequence is an **even function** ($r_{xx}(l) = r_{xx}(-l)$) and attains its **maximum value at zero lag** ($l = 0$).
 - **Application:** It is used to identify periodicities in physical signals that may be heavily corrupted by random noise.

[!EXAMPLE] *Example 6.2*

Compute the **autocorrelation** of the signal

$$x(n) = a^n u(n), \quad 0 < a < 1$$

Solution

1. **Identify Signal Type:** $x(n)$ is an infinite-duration signal; therefore, its autocorrelation sequence $r_{xx}(l)$ will also have **infinite duration**.
2. **Remark:** We must split the solution into two cases, $l > 0$ and $l < 0$, because the **starting point of the overlap** between the signal $x(n)$ and its shifted version $x(n-l)$ changes based on the direction of the shift. Since the signal $x(n) = a^n u(n)$ is **causal** (it is zero for $n < 0$), the product in the autocorrelation formula is only non-zero where both sequences are non-zero:
 - **Case 1** ($l \geq 0$): Shifting $x(n)$ to the right by l units means it starts at $n = l$. The overlap with the original $x(n)$ (which starts at $n = 0$) begins at $n = l$. Thus, the summation limits are from l to ∞ .
 - **Case 2** ($l < 0$): Shifting $x(n)$ to the left by l units means it starts at $n = l$ (a negative value). The overlap with the original $x(n)$ still begins at the original starting point, $n = 0$. Thus, the summation limits are from 0 to ∞ .

3. **Evaluate for $l > 0$:**

- Apply the autocorrelation formula:

$$r_{xx}(l) = \sum_{n=l}^{\infty} x(n)x(n-l)$$

- Substitute the signal:

$$r_{xx}(l) = \sum_{n=l}^{\infty} a^n a^{n-l} = a^{-l} \sum_{n=l}^{\infty} (a^2)^n$$

- Change variables ($k = n - l$) and apply the geometric series formula:

$$r_{xx}(l) = \frac{1}{1-a^2} a^l, \quad l > 0$$

4. **Evaluate for $l < 0$:**

- Apply the formula:

$$r_{xx}(l) = \sum_{n=0}^{\infty} x(n)x(n-l) = a^{-l} \sum_{n=0}^{\infty} (a^2)^n$$

- Apply the geometric series formula:

$$r_{xx}(l) = \frac{1}{1-a^2} a^{-l}, \quad l < 0$$

5. **Generalize the Result:** Since $a^{-l} = a^{|l|}$ when l is negative, the two results can be combined into a single expression for all l :

$$r_{xx}(l) = \frac{1}{1-a^2} a^{|l|}, \quad -\infty < l < \infty$$

The Z-Transform and Its Application to the Analysis of LTI Systems

The Direct Z-Transform

The idea of the **z-transform** is to transform the time-domain signal $x(n)$ into the z-domain, which does not depend on n . It is defined as the following power series:

$$X(z) \equiv \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

where z is a complex variable. The inverse procedure of obtaining $x(n)$ from $X(z)$ is called the **inverse z-transform**.

Region of Convergence (ROC): The ROC is the set of all values of z for which $X(z)$ attains a finite value. Any time a z-transform is cited, its ROC should also be indicated, as a discrete-time signal is only **uniquely determined** by the combination of its z-transform $X(z)$ and its ROC.

[!EXAMPLE] Calculate the z-transform of these signals:

- **Unit Sample Signal** $x(n) = \delta[n]$:

- **Calculation:** Since $\delta(n)$ is 1 only at $n = 0$ and 0 elsewhere:

$$X(z) = \sum_{n=-\infty}^{\infty} \delta(n)z^{-n} = \delta(0)z^{-0} = 1 \cdot 1 = 1$$

- **Result:** $X(z) = 1$

- **ROC:** Entire z-plane.

- **Unit Step Signal** $x(n) = u[n]$:

- **Calculation:** Since $u(n) = 1$ for $n \geq 0$:

$$X(z) = \sum_{n=0}^{\infty} (1)z^{-n} = \sum_{n=0}^{\infty} (z^{-1})^n$$

Applying the Geometric Series property $\sum_{n=0}^{\infty} A^n = \frac{1}{1-A}$ where $A = z^{-1}$:

$$X(z) = \frac{1}{1 - z^{-1}}$$

- **ROC:** Converges if $|z^{-1}| < 1 \Rightarrow |z| > 1$.

- **Exponential Signal** $x(n) = \alpha^n u(n)$:

- **Calculation:**

$$X(z) = \sum_{n=0}^{\infty} \alpha^n z^{-n} = \sum_{n=0}^{\infty} (\alpha z^{-1})^n$$

Applying the Geometric Series property $\frac{1}{1-A}$ where $A = \alpha z^{-1}$:

$$X(z) = \frac{1}{1 - \alpha z^{-1}}$$

- **ROC:** Converges if $|\alpha z^{-1}| < 1 \Rightarrow |z| > |\alpha|$.

- **Anticausal Exponential Signal** $x(n) = -\alpha^n u(-n-1)$:

- **Calculation:** This signal is non-zero only for $n \leq -1$. Let $l = -n$:

$$X(z) = \sum_{n=-\infty}^{-1} (-\alpha^n)z^{-n} = -\sum_{l=1}^{\infty} (\alpha^{-1}z)^l$$

Applying the Geometric Series property $\sum_{l=1}^{\infty} A^l = \frac{A}{1-A}$ where $A = \alpha^{-1}z$:

$$X(z) = -\frac{\alpha^{-1}z}{1 - \alpha^{-1}z} = \frac{1}{1 - \alpha z^{-1}}$$

- **ROC:** Converges if $|\alpha^{-1}z| < 1 \Rightarrow |z| < |\alpha|$.

Key properties:

- **Linearity:** The transform of a linear combination of signals is the same linear combination of their transforms.
- **Time Shifting:** Delaying a signal by k samples corresponds to multiplying its z -transform by z^{-k} .
- **Convolution:** Convolution in the time domain is equivalent to **multiplication** in the z -domain: $Y(z) = H(z)X(z)$. This property greatly simplifies the analysis of LTI systems.

[!NOTE] *Table 1 Characteristic Families of Signals with Their Corresponding ROCs*

Finite-Duration Signals

For signals that exist only over a specific, finite interval of time, the ROC is generally the **entire z -plane**, with specific points excluded depending on where the signal's non-zero samples are located.

- **Causal:** The signal is zero for $n < 0$.
 - **ROC:** Entire z -plane except $z = 0$. The origin is excluded because terms with z^{-k} (where $k > 0$) become unbounded at $z = 0$.
- **Anticausal:** The signal is zero for $n > 0$.
 - **ROC:** Entire z -plane except $z = \infty$. $z = \infty$ is excluded because terms with z^k (where $k > 0$) become unbounded there.
- **Two-sided:** The signal has non-zero samples for both positive and negative values of n .
 - **ROC:** Entire z -plane except $z = 0$ **and** $z = \infty$.

Infinite-Duration Signals

For signals that continue indefinitely in one or both directions, the ROC takes the form of circular regions or rings.

- **Causal:** These signals are zero for $n < 0$ and continue to $n = \infty$.
 - **ROC:** The **exterior of a circle** of radius r_2 ($|z| > r_2$).
- **Anticausal:** These signals are zero for $n \geq 0$ and continue to $n = -\infty$.
 - **ROC:** The **interior of a circle** of radius r_1 ($|z| < r_1$).
- **Two-sided:** These signals have infinite duration in both the positive and negative directions.
 - **ROC:** An **annular region (ring)** in the z -plane defined by $r_2 < |z| < r_1$. If $r_2 > r_1$, the two components' convergence regions do not overlap, and the z -transform does not exist for that signal.

[!NOTE] *Table 2 Properties of the z -Transform*

Fundamental Properties

- **Linearity:** The z -transform of a linear combination of signals is the same linear combination of their individual transforms: $a_1x_1(n) + a_2x_2(n) \leftrightarrow a_1X_1(z) + a_2X_2(z)$. The resulting ROC is at least the **intersection** of the individual ROCs.
- **Time Shifting:** Delaying or advancing a signal by k samples corresponds to multiplying $X(z)$ by z^{-k} : $x(n - k) \leftrightarrow z^{-k}X(z)$. This is a critical property for solving **difference equations**.

- **Scaling in the z-Domain:** Multiplying a signal by an exponential sequence a^n results in scaling the z variable: $a^n x(n) \leftrightarrow X(a^{-1}z)$. This results in **expanding or shrinking** the z -plane and its ROC by a factor of $|a|$.
- **Time Reversal:** Folding a signal $x(n)$ to $x(-n)$ corresponds to replacing z with z^{-1} : $x(-n) \leftrightarrow X(z^{-1})$. The ROC is inverted ($1/r_1 < |z| < 1/r_2$).

Signal Manipulation Properties

- **Conjugation:** For complex signals, $x^*(n) \leftrightarrow X^*(z^*)$.
- **Real and Imaginary Parts:** $\text{Re}\{x(n)\} \leftrightarrow \frac{1}{2}[X(z) + X^*(z^*)]$ and $\text{Im}\{x(n)\} \leftrightarrow \frac{1}{2j}[X(z) - X^*(z^*)]$.
- **Differentiation in the z-Domain:** Multiplying a signal by n is equivalent to differentiating its transform and multiplying by $-z$: $nx(n) \leftrightarrow -z \frac{dX(z)}{dz}$. This is useful for finding transforms of **ramp sequences**.

System and Energy Properties

- **Convolution:** This is arguably the most powerful property: convolution in the time domain is equivalent to **multiplication** in the z -domain: $x_1(n) * x_2(n) \leftrightarrow X_1(z)X_2(z)$. This converts complex time-domain operations into simple algebraic ones.
- **Correlation:** The z -transform of the crosscorrelation $r_{x_1 x_2}(l)$ is $X_1(z)X_2(z^{-1})$.
- **Multiplication of Two Sequences:** Multiplying two signals in time corresponds to a **complex contour integral** (periodic convolution) in the z -domain.
- **Parseval's Relation:** Relates the total energy of signals in the time domain to their representation in the z -domain via a contour integral.

Initial Value Theorem

If a signal $x(n)$ is **causal** (zero for $n < 0$), the first sample value $x(0)$ can be found directly from $X(z)$ by taking the limit as z approaches infinity: $x(0) = \lim_{z \rightarrow \infty} X(z)$.

[!NOTE] *Table 3 Some Common z-Transform Pairs*

Elementary Basic Signals

- **Unit Sample $\delta(n)$:** The transform is simply 1 for the entire z -plane. This is because the signal is only non-zero at $n = 0$.
- **Unit Step $u(n)$:** Expressed as $\frac{1}{1-z^{-1}}$ with a Region of Convergence (ROC) of $|z| > 1$. This represents the sum of a geometric series for a causal signal.

Exponential and Ramp Signals

A critical takeaway from this section is that different time-domain signals can result in the same algebraic $X(z)$ expression; they are only distinguished by their **ROC**.

- **Causal Exponential $a^n u(n)$:** $X(z) = \frac{1}{1-az^{-1}}$ with ROC $|z| > |a|$.
- **Anticausal Exponential $-a^n u(-n-1)$:** Shares the **same algebraic form** ($\frac{1}{1-az^{-1}}$) but has an ROC of $|z| < |a|$.
- **Ramp-like Signals ($na^n u(n)$ and $-na^n u(-n-1)$):** These result in $\frac{az^{-1}}{(1-az^{-1})^2}$. These are derived using the **differentiation property** (multiplying a signal by

n in time corresponds to differentiation in the z -domain).

Sinusoidal and Weighted Sinusoidal Signals

These pairs represent oscillatory behavior and are derived using **Euler's identity** to split the sine and cosine functions into complex exponentials.

- **Standard Sine and Cosine:**

- $\cos(\omega_0 n)u(n) \leftrightarrow \frac{1-z^{-1} \cos \omega_0}{1-2z^{-1} \cos \omega_0 + z^{-2}}.$
- $\sin(\omega_0 n)u(n) \leftrightarrow \frac{z^{-1} \sin \omega_0}{1-2z^{-1} \cos \omega_0 + z^{-2}}.$
- Both have an ROC of $|z| > 1$.

- **Exponentially Weighted Sine and Cosine ($a^n \cos \dots$ and $a^n \sin \dots$):**

- These follow the same structure as the standard sinusoids but include the scaling factor a .
- Their ROC is $|z| > |a|$, illustrating the **Scaling property** where multiplying by a^n in time scales the z variable.